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United States Patent Application Publication

20250277382

Kind Code

A1

Publication Date

September 04, 2025

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REBAR TYING TOOL

Abstract

A rebar tying tool ties together two or more rebars using a wire. The rebar tying tool may include: a reel having the wire wound on a bobbin; a reel-holding part configured to hold the reel in a rotatable manner; a feed part configured to advance the wire from the reel by a number of turns of the wire around the rebars; and a twisting part configured to twist the wire after the number of turns of the wire has been wound around the rebars. The wire preferably has a maximum tensile load of at least 1000 N and a yield-point load of at least 550 N.

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Family ID: 88327582

Appl. No.: 19/214144

Filed: May 21, 2025

Foreign Application Priority Data

JP	2022-061997	Apr. 01, 2022
JP	2022-107137	Jul. 01, 2022

Related U.S. Application Data

parent US continuation 18179483 20230307 parent-grant-document US 12338641 child US 19214144

Publication Classification

Int. Cl.: E04G21/12 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Japanese patent application no. 2022-061997 filed on Apr. 1, 2022, and to Japanese patent application no. 2022-107137 filed on Jul. 1, 2022, the contents of which are fully incorporated herein by reference.

TECHNICAL FIELD

[0002] The disclosure of the present specification relates to a rebar tying tool, e.g., for use in tying together (binding) reinforcing bars used in reinforced concrete.

BACKGROUND ART

[0003] US 2018/207709 discloses a rebar tying tool (binding machine) that ties together (binds) reinforcing bars (hereinafter “rebars”) using a wire. This rebar tying tool comprises: a reel comprising a bobbin and the wire, which is wound around the bobbin; a reel-holding part configured to hold the reel in a rotatable manner; a feeding part (feeding unit) configured to advance the wire from the reel around the rebars; and a twisting part (twisting unit) configured to twist the wire after it has been curled (wound, looped) around the rebars. The wire used with this known rebar tying tool is a steel wire having a diameter of approximately 1 mm.

SUMMARY OF THE INVENTION

[0004] It is one non-limiting object of the present teachings to disclose techniques for tying together rebars more tightly.

[0005] Embodiments of rebar tying tools disclosed herein are configured to tie together (bind) rebars using a wire, e.g., for use in reinforcing concrete. Such rebar tying tools may comprise: a reel comprising a bobbin and the wire, which is wound on the bobbin; a reel-holding part configured to hold the reel in a rotatable manner; a feed part (feed mechanism) configured to advance the wire from the reel around the rebars (preferably by a predetermine number of turns or windings around the rebar; and a twisting part (twisting mechanism) configured to twist (terminal ends of) the wire that is around the rebars (i.e. after the wire has been looped (wound) around two or more rebars).

[0006] In one aspect of the present disclosure, the overall (total) maximum tensile load of the wire looped around the rebars (i.e. the wire wound on the bobbin) may be 1,050 N or more. The overall (total) maximum tensile load is calculated by multiplying the maximum tensile load of the wire by the number of turns of the wire around the rebars.

[0007] To tightly tie together (bind) the rebars, it is necessary to twist terminal ends of the wire that is curled or looped around two or more rebars with a strong force. However, if the overall maximum tensile load of the wire, which is looped around the rebars, is small (low), when the terminal ends of the looped wire are twisted with a strong force, there is a risk that the wire will fracture (break). On the other hand, if the overall maximum tensile load of the wire around the rebars is 1,050 N or more, even if the terminal ends of the wire that is looped around the rebars are twisted with a strong force, it is possible to reduce the likelihood of fracturing of the wire. Thus, by utilizing a wire that will have an overall maximum tensile load of at least 1,050 N after the predetermined number of turns of the wire have been wound around the rebars, the rebars can be tied more tightly.

[0008] In addition or in the alternative, in another aspect of the present teachings, the maximum tensile load per single wire looped around the rebars (i.e. the wire wound on the bobbin) may be 700 N or more.

[0009] As was mentioned above, to tightly tie together the rebars, it is necessary to twist the

terminal ends of the wire that is curled or looped around the rebars with a strong force. However, if the maximum tensile load per single (each) wire looped or curled around the rebars (i.e. the wire wound on the bobbin) is small (low), when the terminal ends of the single looped wire are twisted with a strong force, there is a risk that the wire will fracture (break). On the other hand, if the maximum tensile load per single wire around the rebars is 700 N or more (i.e. if the wire wound on the bobbin has a maximum tensile load of at least 700 N), even if the terminal ends of the wire that is looped around the rebars are twisted with a strong force, it is possible to reduce the likelihood of fracturing of the wire. Thus, if a wire having a maximum tensile load of at least 700 N is wound on the bobbin and used to tie together the rebars, the rebars can be tied together more tightly.

[0010] In addition or in the alternative, in another aspect of the present teachings, the overall (total) yield-point load of the wire looped around the rebars (i.e. the wire wound on the bobbin) may be 700 N or more. The overall (total) yield-point load is calculated by multiplying the yield-point load of the wire by the number of turns of the wire around the rebars.

[0011] If the wire that ties together the rebars deforms in an adverse manner during the advancing or twisting process, one or more gaps might form between the rebars and the wire, and thereby the tying of the rebars will loosen in an adverse manner. Consequently, to tightly tie together the rebars, it is necessary to reduce the likelihood that the wire that ties together the rebars will adversely deform during the advancing or twisting process. If the overall yield-point load of the wire looped around the rebars by the predetermined number of turns is small (low) and a strong force acts on the terminal ends of the looped wire, there is a risk that the wire will deform in an adverse manner. On the other hand, if the overall yield-point load of the wire looped around the rebars by the predetermined number of turns is 700 N or more, even if a strong force acts on the wire that ties the rebars, it is possible to reduce the likelihood of adverse deformation of the wire. In this aspect as well, the rebars can be tied together more tightly.

[0012] In addition or in the alternative, in another aspect of the present teachings, the yield-point load per single wire looped around the rebars (i.e. the wire wound on the bobbin) may be 450 N or more.

[0013] As was mentioned above, if the wire that ties together the rebars deforms in an adverse manner during the advancing or twisting process, one or more gaps might form between the rebars and the wire, and therefore the tying of the rebars will loosen in an adverse manner. Consequently, to tie the rebars tightly, it is necessary to reduce the likelihood that the wire that ties together the rebars will adversely deform during the advancing or twisting process. If the yield-point load per single wire looped around the rebars (i.e. the wire wound on the bobbin) is small (low) and a strong force acts on the looped wire, then there is a risk that the wire will deform in an adverse manner. On the other hand, if the yield-point load per single wire looped around the rebars (i.e. the wire wound on the bobbin) is 450 N or more, even if a strong force acts on the wire that ties together the rebars, it is possible to reduce the likelihood of adverse deformation of the wire. In this aspect as well, the rebars can be tied more tightly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is an oblique view, viewed from the rear, the left, and above, of a rebar tying tool 2 according to a working example.

[0015] FIG. 2 is an oblique view, viewed from the front, the right, and above, of the rebar tying tool 2 according to the working example.

[0016] FIG. 3 is a side view that shows the internal configuration of the rebar tying tool 2 according to the working example.

[0017] FIG. 4 is an oblique view of a feed part 38 according to the working example.

[0018] FIG. 5 is an oblique view of the feed part **38** and a reel holder **10** according to the working example.

[0019] FIG. 6 is a cross-sectional view of the vicinity of a front-side upper portion of the rebar tying tool **2** according to the working example.

[0020] FIG. 7 is a side view that shows, in a cutting part **44** according to the working example, the state before a first lever member **76** and a second lever member **78** pivot.

[0021] FIG. 8 is a side view that shows, in the cutting part **44** according to the working example, the state after the first lever member **76** and the second lever member **78** have pivoted.

[0022] FIG. 9 is an oblique view of a twisting part **46** according to the working example.

[0023] FIG. 10 is a cross-sectional view of a twisting motor **86**, a speed-reducing part **88**, and a holding part **90** according to the working example.

[0024] FIG. 11 is an exploded, oblique view of a carrier sleeve **98**, a clutch plate **100**, and a screw shaft **102** according to the working example.

[0025] FIG. 12 is an oblique view of a clamp shaft **110** according to the working example.

[0026] FIG. 13 is an oblique view of the twisting part **46** according to the working example in the state in which a right clamp **112** and a left clamp **114** are mounted on the clamp shaft **110**.

[0027] FIG. 14 is an oblique view of the right clamp **112** according to the working example.

[0028] FIG. 15 is an oblique view of the left clamp **114** according to the working example.

[0029] FIG. 16 is an oblique view of the twisting motor **86**, the speed-reducing part **88**, and the holding part **90** according to the working example.

[0030] FIG. 17 is an oblique view of the rotation-blocking part **92** according to the working example.

[0031] FIG. 18 is a table that concerns wires W used by a rebar tying tool according to a comparative example.

[0032] FIG. 19 is a table that concerns wires W used by the rebar tying tool **2** according to the working example.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0033] Representative, non-limiting concrete examples of the present invention are explained in detail below, with reference to the drawings. This detailed explanation is intended merely to illustrate to a person skilled in the art that details to implement preferred examples of the present invention are not intended to limit the scope of the present invention. In addition, additional features and the invention disclosed below can be used separately from or together with other features and inventions to provide additional improved rebar tying tools, manufacturing methods, and methods of use.

[0034] In addition, the combinations of features and processes disclosed in the detailed explanation below are not essential to carry out the present invention in the broadest meaning and are recited only to explain representative concrete examples of the present invention in particular.

Furthermore, in providing additional and useful embodiments of the present invention, the various features of the representative concrete examples above and below and the various features of the independent and dependent claims do not necessarily have to be combined as indicated in the concrete examples recited herein or as indicated in the sequence enumerated herein.

[0035] All features recited in the present specification and/or in the patent claims are intended, separately from the configuration of features recited in the working examples and/or the claims, to be disclosed individually and mutually independently as limitations relative to the specific matters disclosed in the disclosure and claims of the original patent application. Furthermore, description related to all numerical ranges, groups, and collections are intended to disclose intermediate configurations thereof as limitations relative to specific matters recited in the disclosure and the claims of the original patent application.

[0036] As was mentioned above, embodiments of rebar tying tools disclosed herein are configured tie together (bind) rebars using a wire, e.g., for use in reinforcing concrete. Such rebar tying tools

may preferably comprise one or more of: a reel comprising a bobbin and the wire, which is wound on the bobbin; a reel-holding part configured to hold the reel in a rotatable manner; a feed part (feed mechanism) configured to advance the wire from the reel by a predetermined number of turns around the rebars; and a twisting part (twisting mechanism) configured to twist (terminal ends of) the wire after the predetermined number of turns have been wound around the rebars (i.e. after the wire has been looped around two or more rebars). Preferably, the overall (total) maximum tensile load of the wire looped around the rebars (i.e. the wire wound on the bobbin) may be 1,050 N or more, wherein the overall (total) maximum tensile load of the wire is calculated by multiplying the maximum tensile load of the wire by the predetermined number of turns of the wire that have been wound around the rebars. For example, the predetermined number of turns of the wire are preferably one or two, although the predetermined number of turns may be three or greater.

[0037] In one or more embodiments, the overall maximum tensile load of the wire after having been wound (looped) around the rebars by the predetermined number of turns may be within the range of 1,050-4,700 N.

[0038] Generally speaking, to increase the overall maximum tensile load of the wire around the rebars, it is necessary to increase the number (i.e. the number of turns or winding count) and/or the diameter of the wires. However, if the number of turns and/or the diameter of the wires is/are excessively large, then there is a risk that an excessive load will act on the motive-power source (e.g., an electric motor) that drives the twisting part. By utilizing a wire such that an overall maximum tensile load of at least 1,050 N will be achieved when the wire has been wound around the rebars by the predetermined number of turns, the rebars can be tied together more tightly (without breaking the wire) while reducing the likelihood that an excessive load will act on the motive-power source that drives the twisting part.

[0039] Preferably, the overall yield-point load of the wire wound (looped) around the rebars after the wire has been wound around the rebars by the predetermined number of turns may be 700 N or more, wherein the overall (total) yield-point load of the wire is calculated by multiplying the maximum yield-point load of the wire by the predetermined number of turns of the wire that have been wound around the rebars. For example, the predetermined number of turns of the wire are preferably one or two, although the predetermined number of turns may be three or greater.

[0040] In one or more embodiments, the overall yield-point load of the wire wound (looped) around the rebars after the wire has been wound the predetermined number of times around the rebars may be within the range of 700-2,550 N.

[0041] Generally speaking, when the overall yield-point load of the wire around the rebars is excessively large, the wire tends not to tightly contact the outer-circumferential surfaces of the rebars after the advancing and/or twisting operations, and thereby one or more gaps can adversely form between the wire and the rebars. In this case, it becomes difficult to tightly tie together the rebars. However, by suitably selecting the overall yield-point load of the wire within the above-noted range (e.g., by appropriately selecting the yield-point load of the wire wound on the bobbin and the predetermined number of turns of the wire around the rebars), the rebars can be tied together more tightly.

[0042] In one or more embodiments, the twisting part (mechanism) may comprise a tip-holding part configured to hold a tip of the wire that has advanced around the rebars. The feed part (mechanism) may be configured to draw back (pull, tension) the wire after the tip-holding part holds the tip of the wire and before the twisting part twists the wire (i.e. the terminal ends of the wire looped around two or more rebars).

[0043] According to the above-mentioned configuration, because the feed part draws back the wire, when the twisting part twists the wire, the wire can be twisted from a state in which the wire has been brought into tighter contact with the rebars. By employing such a configuration, the rebars can be tied together more tightly.

[0044] In one or more embodiments, the diameter of the wire may be 1.6 mm or more. Although

the upper limit of the diameter of the wire is not particularly limited, the diameter of the wire may be, e.g., 5 mm or less, e.g., 4 mm or less when the predetermined number of turns is one and may be, e.g., 2.5 mm or less, e.g., 2.3 mm or less when the predetermined number of turns is two.

[0045] Such a wire enables the rebars to be tied together more tightly.

[0046] In one or more embodiments, the diameter of a trunk part of the bobbin may be 50 mm or more. Although the upper limit of the diameter of the trunk part is not particularly limited, the diameter of the trunk part may be, e.g., 100 mm or less, e.g., 80 mm or less.

[0047] By making the diameter of the trunk part of the bobbin relatively large (i.e. at least 50 mm), it is possible to reduce the likelihood that a winding curl (curved plastic deformation) will be imparted to the wire due to the wire being wound on the bobbin.

[0048] In one or more embodiments, the feed part may comprise feed rollers, which advance the wire by rotating. The hardness of the feed rollers may be 56 HRC or more. Although the upper limit of the hardness of the feed rollers is not particularly limited, the hardness of the feed rollers may be, e.g., 70 HRC or less, e.g., 68 HRC or less.

[0049] According to the above-mentioned configuration, it is possible to reduce the likelihood and/or amount of adverse wear on the feed rollers when the feed part advances the wire.

[0050] In one or more embodiments, the rebar tying tool may further comprise a cutting part comprising cutters configured to cut the wire after the wire has been looped around two or more rebars. The hardness of the cutters may be 56 HRC or more. Although the upper limit of the hardness of the cutters is not particularly limited, the hardness of the cutters may be, e.g., 70 HRC or less, e.g., 68 HRC or less.

[0051] According to the above-mentioned configuration, it is possible to reduce the likelihood and/or amount of adverse wear on the cutters when the cutting part cuts the wire.

[0052] In one or more embodiments, the rebar tying tool may further comprise a grip (handle), which the user grips (grasps, hold) during a rebar tying operation so that the user can perform the work of tying together the rebars in a hand-held state.

[0053] According to the above-mentioned configuration, the rebars can be tied more tightly by using the hand-held-type rebar tying tool.

[0054] Preferably, the maximum tensile load per single wire looped around the rebars (i.e. the wire wound on the bobbin) may be 700 N or more.

[0055] Preferably, the yield-point load per single wire looped around the rebars (i.e. the wire wound on the bobbin) may be 450 N or more.

Working Example

[0056] The rebar tying tool **2** shown in FIG. **1** is configured to tie together (bind) a plurality of rebars (reinforcing bars) **R** using a wire **W**. The rebar tying tool **2** comprises a main body **4**, a grip (handle) **6**, a battery-mount part **8**, a battery pack (battery cartridge) **B**, and a reel holder **10**. The grip **6** is configured to be gripped by the user. The grip **6** is disposed at a rear-side lower portion of the main body **4**. The grip **6** is formed integrally with the main body **4**. A trigger **12** is provided at a front-surface upper portion of the grip **6**. A trigger switch **14** (refer to FIG. **3**), which detects whether the trigger **12** is being pulled, is disposed in the interior of the grip **6**. The battery-mount part **8** is disposed at a lower portion of the grip **6**. The battery-mount part **8** is formed integrally with the grip **6**. The battery pack **B** is detachable by being slid relative to the battery-mount part **8**. The battery pack **B** preferably comprises a secondary (rechargeable) battery such as, for example, one or more lithium-ion battery cells. The reel holder **10** is disposed at a front-side lower portion of the main body **4**. The reel holder **10** is disposed forward of the grip **6**. It is noted that, in the present working example, a longitudinal direction of a twisting part **46**, which is described below, is called the front-rear direction, a direction orthogonal to the front-rear direction is called an up-down direction, and a direction orthogonal to the front-rear direction and the up-down direction is called a left-right direction, as can be seen in the directional notations in the appended drawings.

[0057] The rebar tying tool **2** comprises a housing **16**. The housing **16** constitutes a portion of a

support part **15**. As shown in FIG. 2, the housing **16** comprises a right housing half **18**, a left housing half **20**, and a motor cover **22**. The right housing half **18** defines the shape of the right-half surfaces of the main body **4**, the grip **6**, and the battery-mount part **8**. The left housing half **20** defines the shape of the left-half surfaces of the main body **4**, the grip **6**, and the battery-mount part **8**. The motor cover **22** is mounted on the outer side of the right housing half **18**. As shown in FIG. 1, an operation-and-display part **24** is disposed at (on) a rear-side upper portion of the left housing half **20**. The operation-and-display part **24** comprises a main power-supply switch (e.g., a button switch) **24a** and a main power-supply LED **24b**. When the main power-supply switch **24a** is manipulated (e.g., pressed) by the user, a main power supply of the rebar tying tool **2** is switched from ON to OFF or vice versa. The main power-supply LED **24b** indicates the ON/OFF state of the main power supply of the rebar tying tool **2**.

[0058] As shown in FIG. 2, the reel holder **10** comprises a holder housing **26**, a main cover **28**, and an auxiliary cover **30**. The holder housing **26** and the auxiliary cover **30** constitute a portion of the support part **15**. The holder housing **26** is fixed to a front-side lower portion of the main body **4** and a front portion of the battery-mount part **8**. The left end of the holder housing **26** is open. The main cover **28** is mounted on the holder housing **26** so as to be rotatable about a pivot shaft **26a** of the holder housing **26** at the lower portion of the holder housing **26**. The main cover **28** is biased in the open direction by a torsion spring **31** (refer to FIG. 3). A closed-state detection sensor (not shown), which detects whether the main cover **28** is in the closed state, is mounted on the holder housing **26**. The auxiliary cover **30** covers the right surface of the holder housing **26**. The auxiliary cover **30** demarcates (bounds) an auxiliary space **30a**, which is between the auxiliary cover **30** and the right surface of the holder housing **26**.

[0059] As shown in FIG. 1, a lock lever **32**, which is for retaining the main cover **28** in the closed state, is disposed at (on) a front-side lower portion of the left housing half **20**. When the lock lever **32** is pivoted, the main cover **28** opens relative to the holder housing **26** owing to the biasing force of the torsion spring **31** (refer to FIG. 3). In the state in which the main cover **28** is closed, a housing space **26b** (refer to FIG. 3) is demarcated (bounded) by the holder housing **26** and the main cover **28**. A reel **33** (refer to FIG. 3), which comprises the wire W in a wound state, is disposed in the housing space **26b**. As shown in FIG. 2, a hole **26c** is formed in the front surface of the holder housing **26**. The user can visually assess (confirm) the remaining amount of the wire W on the reel **33** by looking at the reel **33** through the hole **26c**.

[0060] As shown in FIG. 3, the rebar tying tool **2** comprises a control circuit board **36**. The control circuit board **36** is disposed in the interior of the battery-mount part **8**. The control circuit board **36** is electrically connected by wiring (not shown) respectively to the battery pack B, the trigger switch **14**, and the operation-and-display part **24**. In addition, the control circuit board **36** is electrically connected by wiring (not shown) to the closed-state detection sensor (not shown) mounted on the holder housing **26**.

[0061] The rebar tying tool **2** comprises a feed part (feed mechanism) **38**, a guide part (wire guide) **40**, a cutting part (cutter or cutting mechanism) **44**, and the twisting part (twisting mechanism) **46**. The feed part **38** is disposed in the interior of a front-side lower portion of the main body **4**. The guide part **40** is disposed at a front portion of the main body **4**. The cutting part **44** is disposed in the interior of a lower portion of the main body **4**. The twisting part **46** is disposed in the interior of the main body **4**.

Configuration of Feed Part **38**

[0062] As shown in FIG. 4, the feed part **38** comprises a feed motor (i.e. a motor that supplies motive power for moving (advancing and retracting) the wire W) **50**, a speed-reducing part (gear transmission) **52**, and a feed unit **54**. The feed motor **50** is, for example, a brushless DC motor. The feed motor **50** is disposed rightward of the right housing half **18** (refer to FIG. 2) and is covered by the motor cover **22** (refer to FIG. 2). The feed motor **50** is electrically connected by wiring (not shown) to the control circuit board **36**. The feed motor **50** operates using electric power supplied

from the battery pack B (refer to FIG. 2). The feed part **38** is configured to perform an advancing operation/process (a reeling-out or unreeling operation), which advances (reels out) the wire W to the guide part **40**, and a draw-back operation (a pull back or retraction (reeling in) operation), which draws back (reels in) the wire W from the guide part **40**, as will be further described below. [0063] The speed-reducing part **52** comprises, for example, a planetary-gear mechanism. The speed-reducing part **52** reduces the speed of the rotation of the feed motor **50**. In other words, the speed-reducing part is configured to convert the rotational output from the feed motor **50** into a rotational output having a lower rotational speed but higher torque than the rotational output from the feed motor **50**.

[0064] Referring to FIGS. 4-6, the feed unit **54** comprises a base member **56**, a guide member (e.g., a funnel) **58**, a drive gear **60**, a first feed gear **62**, a second feed gear **64**, a release lever **66**, and a compression spring (biasing member) **68**. The guide member **58** is fixed to the base member **56**. The guide member **58** has a guide hole **58a**. The guide hole **58a** has a tapered shape in which a lower-end portion is wide and an upper-end portion is narrow. The wire W passes through the guide hole **58a**.

[0065] Rotation is transmitted from the speed-reducing part **52** to the drive gear **60**. The first feed gear **62** is supported on (by) the base member **56** in a rotatable manner. The first feed gear **62** meshes with the drive gear **60**. Rotation of the drive gear **60** causes the first feed gear **62** to rotate. The first feed gear **62** has a groove **62a** configured to receive a first circumferential half (semi-circle) of the wire W. The groove **62a** is formed on the outer-circumferential surface of the first feed gear **62** in a direction along the rotational direction of the first feed gear **62**. The second feed gear **64** meshes with the first feed gear **62**. The second feed gear **64** is supported on (by) the release lever **66** in a rotatable manner. The second feed gear **64** has a groove **64a** configured to receive a second circumferential half (semi-circle) of the wire W. The groove **64a** is formed on the outer-circumferential surface of the second feed gear **64** in a direction along the rotational direction of the second feed gear **64**. The release lever **66** is supported on the base member **56** so as to be swingable (pivotable) about a pivot shaft **66a**. The compression spring **68** biases (urges) the release lever **66** relative to the right housing half **18** (refer to FIG. 2) in the direction in which the second feed gear **64** approaches the first feed gear **62**. Therefore, the second feed gear **64** is normally pressed toward (against) the first feed gear **62** by the compression spring **68**. In this state, the wire W is elastically sandwiched (pressed, clamped) between the groove **62a** of the first feed gear **62** and the groove **64a** of the second feed gear **64**. As shown in FIG. 5, when the lock lever **32** pivots in the direction that will release the retention of the main cover **28**, the lower end of the release lever **66** is pressed in by the lock lever **32** and moves toward the right housing half **18**. Thereby, the second feed gear **64** separates (moves away) from the first feed gear **62**. In this state, the user can place the wire W of the reel **33** (refer to FIG. 4) between the groove **62a** of the first feed gear **62** and the groove **64a** of the second feed gear **64**. It is noted that, as shown in FIG. 2, a window **16a**, through which the user can see the location where the first feed gear **62** and the second feed gear **64** mesh, is formed in the front surface of the left housing half **20** and the front surface of the motor cover **22**.

[0066] As shown in FIG. 4, when the feed motor **50** causes the first feed gear **62** to rotate while the wire W is sandwiched (pressed, clamped) between the groove **62a** of the first feed gear **62** and the groove **64a** of the second feed gear **64**, the wire W is moved either forward (wire advancing movement) or rearward (wire retracting (pull back) movement). More specifically, in the present working example, when the rotor of the feed motor **50** forwardly rotates (i.e. the rotor rotates in a first rotational direction), the drive gear **60** rotates in direction D1, which is shown in FIG. 4, and the wire W advances from the reel **33** toward the guide part **40**. On the other hand, when the feed motor **50** reversely rotates (i.e. the rotor rotates in a second rotational direction that is opposite of the first rotational direction), the drive gear **60** rotates in direction D2, which is shown in FIG. 4, and the wire W is drawn back (retracted) from the feed part **38** toward the reel **33**.

Configuration of Guide Part **40**

[0067] As shown in FIG. 6, the guide part **40** comprises an upper-side curl guide **70** and a lower-side curl guide **71**. The upper-side curl guide **70** and the lower-side curl guide **71** are disposed at a front portion of the main body **4**. The lower end of the upper-side curl guide **70** is open downward. Thereby, an upper-side wire passageway **70a** is formed in the upper-side curl guide **70**. The lower-side curl guide **71** is disposed downward of the upper-side curl guide **70**. The upper end of the lower-side curl guide **71** is open upward. Thereby, a lower-side wire passageway **71a** is formed in the lower-side curl guide **71**.

[0068] The wire **W** fed from the feed part **38** (refer to FIG. 4) is fed (advanced) to the upper-side wire passageway **70a**. The wire **W** then passes through the upper-side wire passageway **70a** from the rear side toward the front side. At this time, a downward curl is imparted to the wire **W**, i.e. the wire **W** is bent or curved (curled) into a circular shape or a loop shape, as can be seen, e.g., in FIGS. 1 and 3, owing to contact with the curved surfaces of the upper-side curl guide **70**. The wire **W** that has passed through the upper-side wire passageway **70a** is fed into the lower-side wire passageway **71a**. The wire **W** passes through the lower-side wire passageway **71a** from the front side toward the rear side. Thereby, when the upper-side curl guide **70** and the lower-side curl guide **71** are positioned such that the rebars are disposed within the interior space defined thereby, the wire **W** can be wound (looped) around the rebars **R** by the rebar tying tools **2**. As discussed above and below, the feed part **38** is configured to loop (wind) the wire **W** around the rebars **R** by a predetermined number of turns, such that the number of turns (winding count) of the wire around the rebars may be preferably, one or two, although three or more turns may be utilized in other embodiments of the present teachings.

Configuration of Cutting Part **44**

[0069] As shown in FIG. 7, the cutting part (wire severing mechanism) **44** comprises a fixed-cutter member (fixed cutter or blade) **72**, a movable-cutter member (movable cutter or blade) **74**, a first lever member **76**, a second lever member **78**, a link member (link) **80**, and a torsion spring **82**. As shown in FIG. 6, the fixed-cutter member **72** and the movable-cutter member **74** are disposed along the passageway through which the wire **W** is fed from the feed part **38** toward the guide part **40**. The fixed-cutter member **72** has a hole **72a**, through which the wire **W** passes. The movable-cutter member **74** is supported on the fixed-cutter member **72** so as to be pivotable about and slidable on the fixed-cutter member **72**. The movable-cutter member **74** has a hole **74a** (see FIG. 6), through which the wire **W** passes. As shown in FIG. 6, in the state in which the hole **72a** of the fixed-cutter member **72** and the hole **74a** of the movable-cutter member **74** are in communication (hereinbelow, called the communicating state), the wire **W** can pass through the hole **72a** of the fixed-cutter member **72** and the hole **74a** of the movable-cutter member **74**. Thereafter, when the movable-cutter member **74** pivots in direction **D3** (shown in FIG. 6) relative to the fixed-cutter member **72** (hereinbelow, called the cutting state), the wire **W** is cut (severed) by the fixed-cutter member **72** and the movable-cutter member **74**.

[0070] As shown in FIG. 7, the first lever member **76** and the second lever member **78** are fixed to each other. The first lever member **76** and the second lever member **78** are swingable (pivotable) about axis **RX**. The lower end of the first lever member **76** and the lower end of the second lever member **78** are coupled to the rear end of the link member **80** in a pivotable manner. The front end of the link member **80** is coupled to the lower end of the movable-cutter member **74** in a pivotable manner. The rear end of the link member **80** is biased forward by the torsion spring **82**. When the first lever member **76** and the lower end of the second lever member **78** swing in the forward direction, the link member **80** moves forward, and the fixed-cutter member **72** and the movable-cutter member **74** enter the communicating state. As shown in FIG. 8, when the first lever member **76** and the lower end of second lever member **78** swing in the rearward direction (pivot in the counterclockwise direction in the views shown in FIGS. 7 and 8), the link member **80** moves rearward, and the fixed-cutter member **72** and the movable-cutter member **74** enter the cutting state.

Configuration of Twisting Part **46**

[0071] As shown in FIG. **9**, the twisting part **46** comprises a twisting motor **86** (i.e. a motor that supplies motive power for twisting together the two ends (terminal end portions) of a wire W that has been wound (looped) around the rebars R by the predetermined number of turns and then severed), a speed-reducing part (gear transmission) **88**, a holding part (wire holding or clamping mechanism) **90**, and a rotation-blocking part **92**. The twisting motor **86** is, for example, a brushless DC motor. The twisting motor **86** is fixed to the right housing half **18** (refer to FIG. **1**) and the left housing half **20** (refer to FIG. **1**). The twisting motor **86** is electrically connected to the control circuit board **36** (refer to FIG. **3**) by wiring (not shown). The twisting motor **86** operates using electric power supplied from the battery pack B (refer to FIG. **1**).

[0072] The speed-reducing part **88** is fixed to the right housing half **18** and the left housing half **20**. The speed-reducing part **88** comprises, for example, a planetary-gear mechanism. The speed-reducing part **88** reduces the speed of the rotation of the twisting motor **86**. In other words, the speed-reducing part is configured to convert the rotational output from the twisting motor **86** into a rotational output having a lower rotational speed but higher torque than the rotational output from the twisting motor **86**.

[0073] As shown in FIG. **10**, the holding part **90** comprises a bearing box **96**, a carrier sleeve **98**, a clutch plate **100**, a screw shaft **102**, an inner sleeve **104**, an outer sleeve **106**, a push plate **108**, a clamp shaft **110**, a right clamp **112**, and a left clamp **114**.

[0074] The bearing box **96** is fixed to the speed-reducing part **88**. The bearing box **96** supports the carrier sleeve **98** in a rotatable manner via a bearing **96a**. Rotation from the speed-reducing part **88** is transmitted to the carrier sleeve **98**. When the rotor of the twisting motor **86** forwardly rotates (rotates in a first rotational direction), the carrier sleeve **98** rotates in the direction of a left-hand screw, viewed from the rear side. When the twisting motor **86** reversely rotates (rotates in a second rotational direction that is opposite of the first rotational direction), the carrier sleeve **98** rotates in the direction of a right-hand screw, viewed from the rear side.

[0075] As shown in FIG. **11**, clutch grooves **98a**, which extend in the front-rear direction, are formed on the inner surface of a rear portion of the carrier sleeve **98**. A first wall part **98b** and a second wall part **98c** are formed at the front end of each of the clutch grooves **98a**. The distance from the rear end of the carrier sleeve **98** to the first wall part **98b** in the front-rear direction is smaller than the distance from the rear end of the carrier sleeve **98** to the second wall part **98c** in the front-rear direction. The clutch plate **100** is disposed in the interior of the carrier sleeve **98**. Clutch pieces **100a** corresponding to the clutch grooves **98a** are formed on the clutch plate **100**. The clutch plate **100** is biased rearward relative to the carrier sleeve **98** by a compression spring **116**, which is disposed in the interior of the carrier sleeve **98**. The clutch plate **100** can advance relative to the carrier sleeve **98** to a position at which the clutch pieces **100a** make contact with the first wall part **98b** of each of the clutch grooves **98a**. When the wire W is to be twisted, by rotating the carrier sleeve **98** relative to the clutch plate **100** in the direction of a left-hand screw viewed from the rear side, the clutch plate **100** can advance relative to the carrier sleeve **98** to the position at which the clutch pieces **100a** make contact with the second wall part **98c** of each of the clutch grooves **98a**.

[0076] A rear part **102a** of the screw shaft **102** is inserted into the carrier sleeve **98** from the front side and is fixed to the clutch plate **100**. A flange **102c**, which protrudes radially, is formed between the rear part **102a** and a front part **102b** of the screw shaft **102**. A ball groove **102d**, which has a helical shape, is formed on the outer-circumferential surface of the front part **102b** of the screw shaft **102**. An engaging part **102e**, the diameter of which is smaller than that of the front part **102b**, is formed at the front end of the screw shaft **102**.

[0077] As shown in FIG. **10**, a compression spring **118** is mounted on the front part **102b** of the screw shaft **102**. The front part **102b** of the screw shaft **102** is inserted into the inner sleeve **104** from the rear side. Ball holes **104a**, which hold balls **120**, are formed in the inner sleeve **104**. The

balls **120** mate with the ball groove **102d** of the screw shaft **102**. A flange **104b**, which protrudes radially, is formed at a rear end of the inner sleeve **104**. The inner sleeve **104** is inserted into the outer sleeve **106** from the rear side. The outer sleeve **106** is fixed to the inner sleeve **104**. In the state in which rotation of the outer sleeve **106** is permitted by the rotation-blocking part **92** (refer to FIG. **17**), when the screw shaft **102** rotates, the inner sleeve **104** and the outer sleeve **106** rotate integrally. On the other hand, in the state in which rotation of the outer sleeve **106** is prohibited (blocked) by the rotation-blocking part **92**, when the screw shaft **102** rotates, the inner sleeve **104** and the outer sleeve **106** move in the front-rear direction relative to the screw shaft **102**.

Specifically, when the rotor of the twisting motor **86** forwardly rotates and the screw shaft **102** rotates in the direction of a left-hand screw viewed from the rear side, the inner sleeve **104** and the outer sleeve **106** move forward relative to the screw shaft **102**. On the other hand, when the twisting motor **86** reversely rotates and the screw shaft **102** rotates in the direction of a right-hand screw viewed from the rear side, the inner sleeve **104** and the outer sleeve **106** move rearward relative to the screw shaft **102**. The push plate **108** is disposed between the rear end of the outer sleeve **106** and the flange **104b** of the inner sleeve **104**. Consequently, when the inner sleeve **104** and the outer sleeve **106** move in the front-rear direction, the push plate **108** also moves in the front-rear direction. Slits **106a**, which extend rearward from the front end of the outer sleeve **106**, are formed along a front portion of the outer sleeve **106**.

[0078] The clamp shaft **110** is inserted into the inner sleeve **104** from the front side. The engaging part **102e** of the screw shaft **102** is inserted into the rear end of the clamp shaft **110**. The clamp shaft **110** is fixed to the screw shaft **102**. As shown in FIG. **12**, a flat-plate part **110a** and a flange **110c** are formed on the clamp shaft **110**, and an opening **110b** is formed in the clamp shaft **110**. The flat-plate part **110a** is disposed at a front end of the clamp shaft **110** and has a flat-plate shape along the up-down direction and the front-rear direction. A hole **110d**, with which a pin **122** (refer to FIG. **13**) mates, is formed in the flat-plate part **110a**. The opening **110b** is disposed rearward of the flat-plate part **110a**. The opening **110b** passes through the clamp shaft **110** in the left-right direction and extends in the front-rear direction. The flange **110c** is disposed rearward of the opening **110b** and protrudes in the radial direction.

[0079] As shown in FIG. **13**, the right clamp **112** is mounted on the clamp shaft **110** such that the right clamp **112** passes through the opening **110b** of the clamp shaft **110** from the right side to the left side. The left clamp **114** is disposed downward of the right clamp **112** and is mounted on the clamp shaft **110** such that the left clamp **114** passes through the opening **110b** of the clamp shaft **110** from the left side to the right side.

[0080] As shown in FIG. **14**, the right clamp **112** comprises a base part **112a**, a downward protruding part **112b**, an upward protruding part **112c**, a contact part **112d**, an upper-side guard part **112e**, and a front-side guard part **112f**. The base part **112a** has a flat-plate shape along the front-rear direction and the left-right direction. The downward protruding part **112b** is disposed at a right-end portion of the base part **112a** and protrudes downward from the base part **112a**. The upward protruding part **112c** is disposed at a right-front end of the base part **112a** and protrudes upward from the base part **112a**. The contact part **112d** protrudes leftward from the upper end of the upward protruding part **112c**. The upper-side guard part **112e** protrudes leftward from the upper end of the contact part **112d**. The front-side guard part **112f** protrudes leftward from the upward protruding part **112c** and the front end of the contact part **112d**. Cam holes **112g**, **112h** are formed in the base part **112a**. Each of the cam holes **112g**, **112h** has an elongated shape that extends from the rear end to the front end first forward, then bends and extends rightward and forward, and thereafter bends and extends forward.

[0081] As shown in FIG. **15**, the left clamp **114** comprises a base part **114a**, a pin-holding part **114b**, a downward protruding part **114c**, a contact part **114d**, a rear-side guard part **114e**, and a front-side guard part **114f**. The base part **114a** has a flat-plate shape along the front-rear direction and the left-right direction. The pin-holding part **114b** is disposed at the left-front end of the base

part **114a**, is more upward than the base part **114a**, and holds the pin **122** (refer to FIG. **13**) in a slidable manner. The downward protruding part **114c** is disposed at the left-front end of the base part **114a** and protrudes downward from the base part **114a**. The contact part **114d** protrudes rightward from the lower end of the downward protruding part **114c**. The rear-side guard part **114e** protrudes rightward from the rear end of the contact part **114d**. The front-side guard part **114f** protrudes rightward from the front end of the contact part **114d**. Cam holes **114g**, **114h** are formed in the base part **114a**. Each of the cam holes **114g**, **114h** has an elongated shape that extends from the rear end to the front end first forward, then bends and extends leftward and forward, then bends and extends forward, and further bends and extends leftward and forward, after which it bends and extends forward.

[0082] As shown in FIG. **13**, in the state in which the right clamp **112** and the left clamp **114** are mounted on the clamp shaft **110**, a first cam sleeve **124** is disposed such that it passes through the cam hole **112g** and the cam hole **114g**, and a second cam sleeve **126** is disposed such that it passes through the cam hole **112h** and the cam hole **114h**. In addition, a first support pin **128** is disposed such that it passes through the first cam sleeve **124**, and a second support pin **130** is disposed such that it passes through the second cam sleeve **126**. A cushion **131**, which has a circular-ring shape, is mounted between the right clamp **112** and the left clamp **114** on one side and the flange **110c** of the clamp shaft **110** on the other side.

[0083] In the state in which the clamp shaft **110** is mounted on the inner sleeve **104** as shown in FIG. **9**, the right clamp **112** and the left clamp **114** are respectively inserted into the slits **106a** of the outer sleeve **106**, and the first and second support pins **128**, **130** are coupled to the outer sleeve **106**. When the clamp shaft **110** moves in the front-rear direction relative to the outer sleeve **106**, the first cam sleeve **124**, which is mounted on the first support pin **128**, moves in the front-rear direction within the cam holes **112g**, **114g**, and the second cam sleeve **126**, which is mounted on the second support pin **130**, moves in the front-rear direction within the cam holes **112h**, **114h**, and thereby the right clamp **112** and the left clamp **114** move in the left-right direction.

[0084] As shown in FIG. **13**, in an initial state in which the clamp shaft **110** is protruding from the outer sleeve **106** in the forward direction, the right clamp **112** is located at its rightward-most position relative to the left clamp **114**. In this state, a right-side wire passageway **132**, through which the wire **W** passes, is formed between the upward protruding part **112c** of the right clamp **112** and the flat-plate part **110a** of the clamp shaft **110**, and the upper side of the right-side wire passageway **132** is covered by the upper-side guard part **112e**. This state of the right clamp **112** is called the fully open state. From this state, when the outer sleeve **106** moves forward relative to the clamp shaft **110**, the right clamp **112** moves leftward toward the clamp shaft **110**. In this state, the wire **W** is sandwiched between the lower end of the contact part **112d** of the right clamp **112** and the upper end of the flat-plate part **110a** of the clamp shaft **110**, and the front side of the right-side wire passageway **132** is covered by the front-side guard part **112f**. This state of the right clamp **112** is called the fully closed state.

[0085] In the initial state in which the clamp shaft **110** is protruding from the outer sleeve **106** in the forward direction, the left clamp **114** is located at its leftward-most position relative to the clamp shaft **110**. In this state, a left-side wire passageway **134**, through which the wire **W** passes, is formed between the downward protruding part **114c** of the left clamp **114** and the flat-plate part **110a** of the clamp shaft **110**. This state of the left clamp **114** is called the fully open state. From this state, when the outer sleeve **106** moves forward relative to the clamp shaft **110**, the left clamp **114** moves rightward toward the clamp shaft **110**. In this state as well, although the wire **W** can pass through the left-side wire passageway **134**, the rear side of the left-side wire passageway **134** is covered by the rear-side guard part **114e**, and the front side of the left-side wire passageway **134** is covered by the front-side guard part **114f**. This state of the left clamp **114** is called the semi-open state. From this state, when the outer sleeve **106** moves further forward relative to the clamp shaft **110**, the left clamp **114** moves further rightward toward the clamp shaft **110**. In this state, the wire

W is sandwiched (interposed) between the upper end of the contact part **114d** of the left clamp **114** and the lower end of the flat-plate part **110a** of the clamp shaft **110**. This state of the left clamp **114** is called the fully closed state.

[0086] The wire W fed from the feed part **38** (refer to FIG. 6) to the guide part **40** (refer to FIG. 6) passes through the left-side wire passageway **134** before reaching the guide part **40**. Consequently, when the left clamp **114** reaches (assumes) the fully closed state and the wire W is cut by the cutting part **44** (refer to FIG. 6), the terminal end of the wire W wound around the rebars R is held by the left clamp **114** and the clamp shaft **110**.

[0087] In addition, the wire W, which is guided by the guide part **40**, passes through the right-side wire passageway **132**. Consequently, when the right clamp **112** reaches (assumes) the fully closed state, the tip of the wire W wound around the rebars R is held by the right clamp **112** and the clamp shaft **110**.

[0088] As shown in FIG. 16, fins **138** are formed on the outer surface of a rear portion of the outer sleeve **106**. The fins **138** extend in the front-rear direction. In the present working example, eight of the fins **138** are disposed on the outer-circumferential surface of the outer sleeve **106** at a spacing of 45° from each other. In addition, in the present working example, the eight fins **138** comprise seven short fins **138a** and one long fin **138b**. The length of the long fin **138b** in the front-rear direction is longer than the length of the short fins **138a** in the front-rear direction. In the front-rear direction, the location of the rear end of the long fin **138b** is identical to the location of the rear end of each of the short fins **138a**. In the front-rear direction, the front end of the long fin **138b** is located more forward than the front end of each of the short fins **138a**.

[0089] The rotation-blocking part **92** is disposed at a location corresponding to the fins **138** of the outer sleeve **106**. The rotation-blocking part **92** cooperates with the fins **138** to permit or prohibit (block) rotation of the outer sleeve **106**. As shown in FIG. 17, the rotation-blocking part **92** comprises a base member **140**, an upper stopper **142**, a lower stopper **144**, and torsion springs **146**, **148**. The base member **140** is fixed to the right housing half **18** (refer to FIG. 1). The upper stopper **142** is supported at an upper portion of the base member **140** in a swingable manner via a pivot shaft **140a**. The upper stopper **142** comprises a blocking piece **142a**. The blocking piece **142a** is located at a lower portion of the upper stopper **142**. The torsion spring **146** biases the blocking piece **142a** in the direction in which the blocking piece **142a** opens outward (i.e., in the direction in which the blocking piece **142a** goes away from the base member **140**). The lower stopper **144** is supported at a lower portion of the base member **140** in a swingable manner via a pivot shaft **140b**. The lower stopper **144** comprises a blocking piece **144a**. The blocking piece **144a** is located at an upper portion of the lower stopper **144**. The rear end of the blocking piece **144a** is disposed forward of the rear end of the blocking piece **142a**. The torsion spring **148** biases the blocking piece **144a** in the direction in which the blocking piece **144a** opens outward (i.e., in the direction in which the blocking piece **144a** goes away from the base member **140**).

[0090] When, relative to the upper stopper **142**, the rotor of the twisting motor **86** (refer to FIG. 10) forwardly rotates and the screw shaft **102** (refer to FIG. 10) rotates in the direction of a left-hand screw viewed from the rear side, rotation of the outer sleeve **106** is prohibited (blocked) by the upper stopper **142** when the fins **138** (refer to FIG. 16) of the outer sleeve **106** make contact with the blocking piece **142a**. On the other hand, when the twisting motor **86** reversely rotates and the screw shaft **102** rotates in the direction of a right-hand screw viewed from the rear side, the fins **138** of the outer sleeve **106** make contact with the blocking piece **142a** and, as is, push in the blocking piece **142a**. In this situation, the upper stopper **142** prohibits (blocks) rotation of the outer sleeve **106**.

[0091] When, relative to the lower stopper **144**, the rotor of the twisting motor **86** forwardly rotates and the screw shaft **102** rotates in the direction of a left-hand screw viewed from the rear side, even if the fins **138** of the outer sleeve **106** make contact with the blocking piece **144a**, they push in, as is, the blocking piece **144a**. In this situation, the lower stopper **144** does not prohibit (block)

rotation of the outer sleeve **106**. On the other hand, when the screw shaft **102** rotates in the direction of a right-hand screw viewed from the rear side, rotation of the outer sleeve **106** is prohibited (blocked) by the lower stopper **144** when the fins **138** of the outer sleeve **106** make contact with the blocking piece **144a**.

[0092] Next, a representative, non-limiting operation of the rebar tying tool **2** shown in FIG. **1** will be explained. When the trigger **12** is manipulated (pressed) by the user, the rebar tying tool **2** performs a tying operation. When the rebar tying tool **2** performs the tying operation, the rebar tying tool **2** performs, e.g., an advancing process (wire advancing process), a tip-holding process (wire tip holding (clamping) process), a draw-back process (wire draw back (retraction) process), a terminal-end holding process (a wire terminal-end holding (clamping) process), a cutting process (wire severing process), a twisting process (wire-ends twisting together process), and a returning process in this order.

Advancing Process

[0093] When the rotor of the feed motor **50** shown in FIG. **4** forwardly rotates (i.e., rotates in direction **D1** shown in FIG. **4**) from the initial state of the rebar tying tool **2**, the feed part **38** advances the wire **W** on the reel **33** by a prescribed length, so that the predetermined number of turns of the wire are wound (looped) around the rebar. This wire advancement (wire unreeling) causes the tip (front-end portion) of the wire **W** to pass through, in order, the fixed-cutter member **72**, the movable-cutter member **74**, the left-side wire passageway **134**, the guide part **40**, and the right-side wire passageway **132**. Thereby, the wire **W** is wound (looped or wrapped) in a circular-ring-like manner or a loop around the rebars **R**. When the advancing of the wire **W** has completed, the feed motor **50** stops.

Tip-Holding Process

[0094] After the advancing process ends, the twisting motor **86** shown in FIG. **10** is energized to cause its rotor to forwardly rotate, whereby the screw shaft **102** is rotated in the direction of a left-hand screw. At this time, the rotation of the outer sleeve **106** in the direction of a left-hand screw is prohibited (block) by the rotation-blocking part **92**. Consequently, the outer sleeve **106**, together with the inner sleeve **104**, advances relative to the clamp shaft **110**, the right clamp **112** reaches the (its) fully closed state, and the left clamp **114** reaches the (its) semi-open state. Thereby, the tip (front-end portion, first terminal end portion) of the wire **W** is held by the right clamp **112** and the clamp shaft **110**. When it is detected that the tip of the wire **W** is held, the twisting motor **86** stops.

Draw-back Process

[0095] After the tip-holding process ends, the feed motor **50** shown in FIG. **4** is energized to cause its rotor to reversely rotate (i.e., rotate in direction **D2** shown in FIG. **4**), whereby the feed part **38** draws back (retracts, pulls back, tensions) the wire **W** wound (looped) around the rebars **R**. Because the tip of the wire **W** is held by the right clamp **112** and the clamp shaft **110**, the circular-ring-like (loop) shape of the wire **W** around the rebars **R** shrinks in diameter. When the drawing back of the wire **W** is completed, the feed motor **50** stops.

Terminal-End Holding Process

[0096] After the draw-back process ends, the twisting motor **86** shown in FIG. **10** is energized to cause its rotor to forwardly rotate, whereby the screw shaft **102** is rotated in the direction of a left-hand screw. At this time, rotation of the outer sleeve **106** in the direction of a left-hand screw is prohibited (blocked) by the rotation-blocking part **92**. Consequently, the outer sleeve **106**, together with the inner sleeve **104**, advances relative to the clamp shaft **110**, and the left clamp **114** reaches the (its) fully closed state. Thereby, the terminal end (rear-end portion, second terminal end portion) of the wire **W** is held by the left clamp **114** and the clamp shaft **110**.

Cutting Process

[0097] After the terminal-end holding process ends, the twisting motor **86** shown in FIG. **10** is energized to cause its rotor to forwardly rotate further, whereby the screw shaft **102** continues to rotate in the direction of a left-hand screw. At this time, rotation of the outer sleeve **106** in the

direction of a left-hand screw is prohibited (blocked) by the rotation-blocking part **92**. Consequently, the outer sleeve **106**, together with the inner sleeve **104**, further advances relative to the clamp shaft **110** and, as shown in FIG. **8**, the push plate **108** pushes the upper end of the second lever member **78** downward and forward. Thereby, the wire W is cut (severed) by the fixed-cutter member **72** and the movable-cutter member **74**. When the cutting of the wire W is complete, the twisting motor **86** stops.

Twisting Process

[0098] After the cutting process ends, the twisting motor **86** shown in FIG. **10** is energized to cause its rotor to forwardly rotate further, whereby the screw shaft **102** continues to rotate in the direction of a left-hand screw. At this time, rotation of the outer sleeve **106** in the direction of a left-hand screw is permitted by the rotation-blocking part **92**. Consequently, the outer sleeve **106**, the inner sleeve **104**, the clamp shaft **110**, the right clamp **112**, and the left clamp **114** become integral and rotate together in the direction of a left-hand screw. Thereby, the end portions (i.e. the tip (front-end portion) and the terminal end (rear-end portion) of the wire W that has been wound around the rebars R are twisted together. When the twisting of the wire W is complete, the twisting motor **86** stops.

Returning Process

[0099] After the cutting process ends or after the twisting process ends, the twisting motor **86** shown in FIG. **10** is energized to cause its rotor to reversely rotate, whereby the screw shaft **102** rotates in the direction of a right-hand screw. At this time, rotation of the outer sleeve **106** in the direction of a right-hand screw is prohibited (blocked) by the rotation-blocking part **92**. Consequently, the outer sleeve **106**, together with the inner sleeve **104**, retracts relative to the clamp shaft **110**. The left clamp **114** transits the semi-open state and reaches the (its) fully open state, and the right clamp **112** reaches the fully open state. Subsequently, when rotation in the direction of a right-hand screw is permitted by the rotation-blocking part **92**, the outer sleeve **106**, the inner sleeve **104**, the clamp shaft **110**, the right clamp **112**, and the left clamp **114** become integral and rotate together in the direction of a right-hand screw. When the long fin **138b** makes contact with the lower stopper **144**, rotation of the outer sleeve **106** is once again prohibited (blocked), and the outer sleeve **106**, together with the inner sleeve **104**, once again retract relative to the clamp shaft **110**. When it is detected that the twisting part **46** has returned to the (its) initial state, the twisting motor **86** stops.

[0100] The rebar tying tool **2** according to the present working example can perform a single-winding-type tying operation, wherein the wire W is wound one turn around the rebars R and the one wire W is twisted. In addition, the rebar tying tool **2** according to the present working example can also perform a double-winding-type tying operation, wherein the wire W is wound two turns around the rebars R and two of the wires W are twisted at the same time, as will be further explained below.

Single-Winding-Type Tying Operation

[0101] When the single-winding-type tying operation is to be performed (i.e. the predetermined number of turns of the wire around the rebars is one), the rebar tying tool **2** performs, in order, the advancing process, the tip-holding process, the draw-back process, the terminal-end holding process, the cutting process, the twisting process, and the returning process. In this single-winding-type tying operation, the wire W is advanced by the feed part **38**, the tip of the wire W is held by the twisting part **46**, the wire W is drawn back by the feed part **38**, the terminal end of the wire W is also held by the twisting part **46**, and the wire W is cut by the cutting part **44**. Thereafter, the first and second terminal end portions of the wire W are twisted together by the twisting part **46**.

Double-Winding-Type Tying Operation

[0102] When the double-winding-type tying operation is performed (i.e. the predetermined number of turns of the wire around the rebars is two), first, the rebar tying tool **2** performs, in order, the advancing process, the tip-holding process, the draw-back process, the terminal-end holding

process, and the cutting process, and then performs the returning process. In this double-winding-type tying operation, the first turn of the wire W is advanced by the feed part **38**, the tip of the first turn of the wire W is held by the twisting part **46**, the first turn of the wire W is drawn back by the feed part **38**, the terminal end of the first turn of the wire W is also held by the twisting part **46**, and the first turn of the wire W is cut by the cutting part **44**. Thereafter, the twisting part **46** releases the holding of the tip and the holding of the rear end of the first turn of the wire W. Subsequently, the rebar tying tool **2** performs, in order, the advancing process, the tip-holding process, the draw-back process, the terminal-end holding process, the cutting process, the twisting process, and an initial-state returning process. At this time, the second turn of the wire W is advanced by the feed part **38**, the tip of the first turn of the wire W and the tip of the second turn of the wire W are held by the twisting part **46**, the second turn of the wire W is drawn back by the feed part **38**, the terminal end portion of the first turn of the wire W and the terminal end portion of the second turn of the wire W are held by the twisting part **46**, and the second turn of the wire W is cut by the cutting part **44**. From this state, the terminal end portions of the first turn of the wire W and the terminal end portions of the second turn of the wire W are twisted together by the twisting part **46**.

Configuration of Reel **33**

[0103] As shown in FIG. **3**, the reel **33** is held by the reel holder **10** in a rotatable manner. As shown in FIG. **4**, the reel **33** comprises a bobbin **160** and the wire W, which is wound on the bobbin **160**. The bobbin **160** comprises a substantially circular-tube-shaped trunk part and substantially disk-shaped flanges, which are disposed on both ends of the trunk part, and the wire W is wound on (around) the trunk part of the bobbin **160**.

[0104] FIG. **18** shows, as a comparative example, the tensile strength, the wire diameter, the cross-sectional area, the winding count, the overall (total) cross-sectional area, the overall (total) maximum tensile load, the yield stress, and the overall (total) yield-point load of the wire W used in a rebar tying tool (not shown) according to pre-existing technology. FIG. **19** shows the tensile strength, the wire diameter, the cross-sectional area, the winding count (number of turns), the overall cross-sectional area, the overall maximum tensile load, the yield stress, and the overall yield-point load of the wire W used in the rebar tying tool **2** according to the present working example. It is noted that, in the present specification, the yield stress of the wire W means the stress at which the strain of the wire W is 0.2% (the so-called 0.2% proof stress), and the (overall) yield-point load of the wire W means the (overall) load at which the strain of the wire W is 0.2%.

[0105] As shown in FIG. **19**, a variety of the wires W can be used in the rebar tying tool **2** of the present working example. For example, the material of the wire W may be annealed steel wire, zinc-plated wire, polyester-cord wire, stainless-steel wire, or the like. If the wire W is composed of annealed steel wire, zinc-plated wire, or polyester-cord wire, the tensile strength of the wire W is, for example, approximately 350 MPa, and the yield stress of the wire W is, for example, approximately 233 MPa. If the wire W is composed of stainless-steel wire, the tensile strength of the wire W is, for example, approximately 510 MPa, and the yield stress of the wire W is, for example, approximately 279 MPa. It is noted that the material of the wire W may be changed to a material having a higher tensile strength and/or yield stress or may be changed to a material having a lower tensile strength and/or yield stress.

[0106] For example, when the rebars R are tied together using the single-winding method of the rebar tying tool **2** (in FIG. **19**, denoted as a Winding Count of 1), the diameter of the wire W (in FIG. **19**, denoted as Wire Diameter) is 1.6 mm or more, or may be for example 2.0 mm or more, or may be for example 2.4 mm or more, or may be for example 2.8 mm or more, or may be for example 3.2 mm or more. It is noted that, when the rebars R are tied together using the single-winding method of the rebar tying tool **2**, the limit of the maximum diameter of the wire W is approximately 5.0 mm. When the rebars R are tied together using the single-winding method of the rebar tying tool **2**, the overall maximum tensile load of the wire W around the rebars R is equal to the maximum tensile load of a single wire W, because the predetermined number of turns of the

wire W around the rebars R is one. If the tensile strength of the wire W is 350 MPa, the overall maximum tensile load of the wire W around the rebars R is 700 N or more, or may be for example 1,050 N or more, or may be for example 1,550 N or more, or may be for example 2,150 N or more, or may be for example 2,800 N or more. If the tensile strength of the wire W is 510 MPa, the overall maximum tensile load of the wire W around the rebars R is 1,000 N or more, or may be for example 1,600 N or more, or may be for example 2,300 N or more, or may be for example 3,100 N or more, or may be for example 4,100 N or more. Thus, by making the overall maximum tensile load of the wire W around the rebars R and/or by making the maximum tensile load per single wire W around the rebars R high, the wire W can be twisted with a stronger force, and thereby the rebars R can be more tightly tied. In addition, when the rebars R are tied using the single-winding method of the rebar tying tool 2, the overall yield-point load of the wire W around the rebars R is equal to the yield-point load of a single wire W, because the predetermined number of turns of the wire W around the rebars R is one. If the yield stress of the wire W is 233 MPa, the overall yield-point load of the wire W around the rebars R is 450 N or more, or may be for example 700 N or more, or may be for example 1,050 N or more, or may be for example 1,400 N or more, or may be for example 1,850 N or more. If the yield stress of the wire W is 279 MPa, the overall yield-point load of the wire W around the rebars R is 550 N or more, or may be for example 850 N or more, or may be for example 1,250 N or more, or may be for example 1,700 N or more, or may be for example 2,200 N or more. When a force that exceeds the overall yield-point load of the wire W around the rebars R acts on the wire W, the wire W that ties together the rebars R may deform in an adverse manner. When the wire W that ties the rebars R deforms in an adverse manner, one or more gaps is (are) formed between the rebars R and the wire W, and thereby the tying of the rebars R loosens in an adverse manner. As described above, by making the overall yield-point load of the wire W around the rebars R and/or the overall yield-point load per single wire W around the rebars R high, it is possible to make it more difficult for the wire W that ties together the rebars R to deform in an adverse manner, and thereby the rebars R can be tied more tightly.

[0107] In addition, when the rebars R are tied using the double-winding method of the rebar tying tool 2 (in FIG. 19, denoted as a Winding Count of 2), the diameter of the wire W (in FIG. 19, denoted as Wire Diameter) is 1.6 mm or more, or may be for example 2.0 mm or more, or may be for example 2.4 mm. It is noted that, when the rebars R are tied using the double-winding method of the rebar tying tool 2, the limit of the maximum diameter of the wire W is approximately 2.5 mm. The overall maximum tensile load of the wire W around the rebars R in this situation is equal to the maximum tensile load of two of the wires W, because the predetermined number of turns of the wire W around the rebars R is two. If the tensile strength of the wire W is 350 MPa, the overall maximum tensile load of the wire W around the rebars R is 1,400 N or more, or may be for example 2,150 N or more, or may be for example 3,150 N or more. If the tensile strength of the wire W is 510 MPa, then the overall maximum tensile load of the wire W around the rebars R is 2,050 N or more, or may be for example 3,200 N or more, or may be for example 4,600 N or more. Thus, by making the overall maximum tensile load of the wire W around the rebars R and/or the maximum tensile load per single wire W around the rebars R high, when the wire W is to be twisted, it is possible to reduce the likelihood of fracturing of the wire W, and therefore the wire W can be twisted with a stronger force, and the rebars R can be tied more tightly. In addition, when the rebars R are tied using the double-winding method of the rebar tying tool 2, the overall yield-point load of the wire W around the rebars R is equal to the yield-point load of two of the wires W, because the predetermined number of turns of the wire W around the rebars R is two. If the yield stress of the wire W is 233 MPa, the overall yield-point load of the wire W around the rebars R is 900 N or more, or may be for example 1,450 N or more, or may be for example 2,100 N or more. If the yield stress of the wire W is 279 MPa, the overall yield-point load of the wire W around the rebars R is 1,100 N or more, or may be for example 1,750 N or more, or may be for example 2,500 N or more. When a force that exceeds the overall yield-point load of the wire W around the rebars

R acts on the wire W, the wire W that ties together the rebars R deforms in an adverse manner. When the wire W that ties the rebars R deforms in an adverse manner, a gap or gaps is (are) formed between the rebars R and the wire W, and thereby the tying (binding) of the rebars R loosens in an adverse manner. As described above, by making the overall yield-point load of the wire W around the rebars R and/or the yield-point load per single wire W around the rebars R high, it is possible to make it more difficult for the wire W that ties together the rebars R to deform in an adverse manner, and thereby the rebars R can be tied more tightly.

[0108] With regard to the rebar tying tool **2** of the present working example, the bobbin **160** having a trunk-part diameter of 50 mm or more, or for example 52 mm or more, or for example 54 mm or more, is used as the bobbin **160** of the reel **33** shown in FIG. **4**. By employing such a configuration, it is possible to reduce the likelihood of fracturing of the wire W caused by winding the wire W on (around) the bobbin **160**. With regard to the rebar tying tool **2** of the present working example, in the guide part **40** shown in FIG. **3**, the spacing between the tip of the upper-side curl guide **70** and the tip of the lower-side curl guide **71** is within the range of, for example, 50-80 mm. If a winding curl (curve plastic deformation) were to be adversely imparted to the wire W due to the wire W being wound on the bobbin **160**, then there would be a risk that, in the guide part **40**, the tip of the wire W advanced from the upper-side wire passageway **70a** of the upper-side curl guide **70** will not enter the lower-side wire passageway **71a** of the lower-side curl guide **71**, and therefore it would no longer be possible to guide the wire W around the rebars R. According to the above-mentioned configuration, in the guide part **40**, the tip of the wire W that has passed through the upper-side wire passageway **70a** of the upper-side curl guide **70** can reliably be fed to the lower-side wire passageway **71a** of the lower-side curl guide **71**.

[0109] With regard to the rebar tying tool **2** according to the present working example, in the feed part **38** shown in FIG. **4**, the biasing force of the compression spring **68** is adjusted such that the second feed gear **64** pushes against the first feed gear **62** with a force of 60 N or more, or for example 80 N or more, or for example 100 N or more, or for example 120 N or more, or for example 140 N or more, or for example 160 N or more, or for example 180 N. By employing such a configuration, even if the diameter of the wire W is large, the advancing and the drawing back of the wire W can be performed reliably.

[0110] With regard to the rebar tying tool **2** according to the present working example, in the feed part **38**, a metal having a hardness of 56 HRC or more, or for example a hardness of 58 HRC or more, or for example a hardness of 60 HRC or more, or for example a hardness of 62 HRC or more, or for example a hardness of 64 HRC or more, or for example a hardness of 66 HRC or more is used as the material of the first feed gear **62** and the second feed gear **64**. By employing such a configuration, even if the diameter of the wire W is large and, attendant therewith, the first feed gear **62** and the second feed gear **64** are strongly pressed against the wire W, wear of the first feed gear **62** and the second feed gear **64** can be curtailed.

[0111] With regard to the rebar tying tool **2** according to the present working example, in the feed part **38**, a motor having a rated output within the range of 100-500 W, or for example within the range of 150-400 W, is used as the feed motor **50**. By employing such a configuration, even if the diameter of the wire W is large, the advancing and drawing back of the wire W can be performed reliably.

[0112] With regard to the rebar tying tool **2** of the present working example, in the guide part **40** shown in FIG. **3**, a metal having a hardness of 56 HRC or more, or for example a hardness of 58 HRC or more, or for example a hardness of 60 HRC or more, or for example a hardness of 62 HRC or more, or for example a hardness of 64 HRC or more, or for example a hardness of 66 HRC or more, is used as the material of the upper-side curl guide **70** and the lower-side curl guide **71**. By employing such a configuration, even if the diameter of the wire W is large, it is possible to better reduce the likelihood and/or amount of wear caused by the wire contacting the upper-side curl guide **70** and the lower-side curl guide **71**, e.g., during the advancing process (i.e. while the wire W

is being plastically deformed into a loop shape.

[0113] With regard to the rebar tying tool **2** of the present working example, in the cutting part **44** shown in FIG. **6**, a metal having a hardness of 56 HRC or more, or for example a hardness of 58 HRC or more, or for example a hardness of 60 HRC or more, or for example a hardness of 62 HRC or more, or for example a hardness of 64 HRC or more, or for example a hardness of 66 HRC or more, is used as the material of the fixed-cutter member **72** and the movable-cutter member **74**. By employing such a configuration, even if the diameter of the wire **W** is large, it is possible to reduce the likelihood and/or amount of wear of the fixed-cutter member **72** and the movable-cutter member **74** caused by repetitively cutting the wire **W**.

[0114] With regard to the rebar tying tool **2** according to the present working example, in the twisting part **46** shown in FIG. **9**, a metal having a hardness of 56 HRC or more, or for example a hardness of 58 HRC or more, or for example a hardness of 60 HRC or more, or for example a hardness of 62 HRC or more, or for example a hardness of 64 HRC or more, or for example a hardness of 66 HRC or more, is used as the material of the clamp shaft **110**, the right clamp **112**, and the left clamp **114**. By employing such a configuration, even if the diameter of the wire **W** is large, it is possible to reduce the likelihood and/or amount of wear of the clamp shaft **110**, the right clamp **112**, and the left clamp **114** caused by the wire **W** sliding relative to (against) the clamp shaft **110**, the right clamp **112**, and the left clamp **114** in the advancing process and the draw-back process.

[0115] With regard to the rebar tying tool **2** according to the present working example, in the twisting part **46**, a motor having a rated output within the range of 100-500 W, or for example within the range of 150-400 W, is used as the twisting motor **86**. By employing such a configuration, even if the diameter of the wire **W** is large, the twisting of the end portions of the wire **W** can be performed reliably, and thereby the wire **W** can be tightly tied.

Modified Examples

[0116] In the tying operation of the rebar tying tool **2** described above, the draw-back process may be omitted for either the single-winding method or the double-winding method.

[0117] In the rebar tying tool **2** described above, the feed motor **50** and/or the twisting motor **86** may be a brushed DC motor, or may be an AC motor, or may be some other type of motor.

[0118] In the rebar tying tool **2** described above, the reel holder **10**, the feed part **38**, the guide part **40**, the cutting part **44**, and the twisting part **46** may be differently arranged. For example, the reel holder **10** may be disposed on a rear-side upper portion of the main body **4**, the feed part **38** may be disposed between the reel holder **10** and the guide part **40** at an upper portion of the main body **4**, or the cutting part **44** may be disposed between the feed part **38** and the guide part **40** in the interior of the main body **4**.

[0119] In the rebar tying tool **2** described above, the right clamp **112** and the clamp shaft **110** may be configured such that they do not hold the tip of the wire **W**, or the left clamp **114** and the clamp shaft **110** may be configured such that they do not hold the terminal end of the wire **W**; and in the tying operation of the rebar tying tool **2**, the tip-holding process and the terminal-end holding process may be omitted. In such a modified embodiment, in the twisting process, the wire **W** that has been wound around the rebars **R** is twisted by being wound in response to the rotation of the clamp shaft **110**, the right clamp **112**, and the left clamp **114**.

[0120] In the rebar tying tool **2** described above, instead of the battery-mount part **8**, on which the battery pack **B** is mountable, a power-supply-cord connecting part, to which a power-supply cord that supplies electric power from an external power supply can connect, may be provided. In this situation, the rebar tying tool **2** operates using electric power supplied via the power-supply cord.

[0121] Instead of the rebar tying tool **2** being used by the user gripping the grip **6**, the rebar tying tool **2** may be used by being installed on a rebar-tying robot that comprises a transport unit, which transports the rebar tying tool **2**, and a manipulation unit, which manipulates the trigger **12**.

Features of the Embodiments

[0122] In one or more of the embodiments as described above, the rebar tying tool **2** ties together the rebars **R** using the wire **W**. The rebar tying tool **2** comprises: the reel **33** comprising the bobbin **160** and the wire **W**, which is wound on the bobbin **160**; the reel-holding part **10** (example of reel-holding part) configured to hold the reel **33** in a rotatable manner; the feed part **38** configured to advance the wire **W** from the reel **33** a predetermined number of turns around the rebars **R**; and the twisting part **46** configured to twist the wire **W** that has been looped around the rebars **R**. The overall maximum tensile load of the wire **W** after the wire **W** has been wound around the rebars **R** by the predetermined number of turns is 1,050 N or more.

[0123] As was described above, to tightly tie together the rebars **R**, it is necessary to twist the wire **W** that is looped around the rebars **R** with a strong force. Nevertheless, if the overall maximum tensile load of the wire **W** around the rebars **R** is small, when the wire **W** that is around the rebars **R** is twisted with a strong force, there is a risk that the wire **W** will fracture. According to the above-mentioned configuration, because the overall maximum tensile load of the wire **W** around the rebars **R** is 1,050 N or more, even if the wire **W** that is around the rebars **R** is twisted with a strong force, it is possible to reduce the likelihood of fracturing of the wire **W**. By employing such a configuration, the rebars **R** can be tied more tightly.

[0124] In one or more embodiments, the overall maximum tensile load of the wire **W** around the rebars **R** is within the range of 1,050-4,700 N.

[0125] Generally speaking, to make the overall maximum tensile load of the wire **W** around the rebars **R** large, it is necessary to make the number and diameter of the wires **W** large. Nevertheless, if the number of turns and/or the diameter of the wires **W** is (are) excessively large, then there is a risk that an excessive load will act on the motive-power source (e.g., the twisting motor **86**) that drives the twisting part **46**. According to the above-mentioned configuration, the rebars **R** can be tied more tightly while reducing the likelihood that an excessive load will act on the motive-power source that drives the twisting part **46** during the twisting operation.

[0126] In one or more embodiments, the rebar tying tool **2** ties together rebars **R** using the wire **W**. The rebar tying tool **2** comprises: the reel **33** comprising the bobbin **160** and the wire **W**, which is wound on the bobbin **160**; the reel-holding part **10** (example of a reel-holding part) configured to hold the reel **33** in a rotatable manner; the feed part **38** configured to advance the wire **W** around the rebars **R** from the reel **33**; and the twisting part **46** configured to twist the wire **W** that is around the rebars **R**. The overall yield-point load of the wire **W** after the wire **W** has been wound around the rebars **R** by the predetermined number of turns is 700 N or more.

[0127] If the wire **W** that ties together the rebars **R** deforms in an adverse manner, one or more gaps may form between the rebars **R** and the wire **W**, and thereby the tying (binding) of the rebars **R** will loosen in an adverse manner. Consequently, to tightly tie the rebars **R** together, it is necessary to make it difficult for the wire **W** that ties together the rebars **R** to deform. Nevertheless, if the overall yield-point load of the wire **W** around the rebars **R** is small, if a strong force acts on the wire **W** that ties together the rebars **R**, there is a risk that the wire **W** will deform in an adverse manner.

According to the above-mentioned configuration, because the overall yield-point load of the wire **W** around the rebars **R** is 700 N or more, even if a strong force acts on the wire **W** that ties the rebars **R**, it is possible to reduce the likelihood of adverse deformation of the wire **W**. By employing such a configuration, the rebars **R** can be tied more tightly.

[0128] In one or more embodiments, the overall yield-point load of the wire **W** around the rebars **R** is within the range of 700-2,550 N.

[0129] Generally speaking, if the overall yield-point load of the wire **W** around the rebars **R** is excessively large, it becomes difficult for the wire **W** to tightly contact the outer-circumferential surfaces of the rebars **R**, and thereby one or more gaps may adversely form between the wire **W** and the rebars **R**, and it becomes difficult to tightly tie together the rebars **R**. According to the above-mentioned configuration, the rebars **R** can be tied more tightly.

[0130] In one or more embodiments, the twisting part **46** comprises the holding part **90** (example of

a tip-holding part) configured to hold the tip of the wire W that has advanced around the rebars R. The feed part **38** is configured to draw back the wire W after the holding part **90** holds the tip of the wire W and before the twisting part **46** twists the wire W.

[0131] According to the above-mentioned configuration, because the feed part **38** draws back the wire W, when the twisting part **46** twists the wire, the wire W can be twisted from a state in which the wire W has been brought into tighter contact with the rebars R. By employing such a configuration, the rebars R can be tied more tightly.

[0132] In one or more embodiments, the diameter of the wire W may be 1.6 mm or more.

[0133] According to the above-mentioned configuration, the rebars R can be tied together more tightly.

[0134] In one or more embodiments, the diameter of the trunk part of the bobbin **160** is 50 mm or more.

[0135] According to the above-mentioned configuration, it is possible to reduce the likelihood of a winding curl being imparted to the wire W due to the wire W being wound (and stored) on the bobbin **160**.

[0136] In one or more embodiments, the feed part **38** comprises the first feed gear **62** and the second feed gear **64** (example of feed rollers), which advances the wire W by rotating. The hardness of the first feed gear **62** and the second feed gear **64** is 56 HRC or more.

[0137] According to the above-mentioned configuration, it is possible to reduce the likelihood and/or amount of adverse wear of the first feed gear **62** and the second feed gear **64** when the feed part **38** advances the wire W.

[0138] In one or more embodiments, the rebar tying tool **2** further comprises the cutting part **44** comprising the fixed-cutter member **72** and the movable-cutter member **74** (example of cutters) that cut the wire W. The hardness of the fixed-cutter member **72** and the movable-cutter member **74** is 56 HRC or more.

[0139] According to the above-mentioned configuration, it is possible to reduce the likelihood and/or amount of adverse wear of the fixed-cutter member **72** and the movable-cutter member **74** when the cutting part **44** cuts the wire W.

[0140] In one or more embodiments, the rebar tying tool **2** further comprises the grip **6**, which the user grips. The user is capable of performing the work of tying the rebars R in the state in which the user grips the rebar tying tool **2** with their hand.

[0141] According to the above-mentioned configuration, the rebars R can be tied together more tightly by using the hand-held type rebar tying tool **2**.

[0142] In one or more embodiments, the rebar tying tool **2** ties together rebars R using the wire W. The rebar tying tool **2** comprises: the reel **33** comprising the bobbin **160** and the wire W, which is wound on the bobbin **160**; the reel holder **10** (example of a reel-holding part) configured to hold the reel **33** in a rotatable manner; the feed part **38** configured to advance the wire W around the rebars R from the reel **33**; and the twisting part **46** configured to twist the wire W that is around the rebars R. The maximum tensile load per single wire W around the rebars R is 700 N or more.

[0143] To tightly tie the rebars R, it is necessary to twist the wire W that is around the rebars R with a strong force. Nevertheless, if the maximum tensile load per single wire W around the rebars R is small, when the wire W that has been around the rebars R is twisted with a strong force, there is a risk that the wire W will fracture. According to the above-mentioned configuration, because the maximum tensile load per single wire W around the rebars R is 700 N or more, even if the wire W that is around the rebars R is twisted with a strong force, it is possible to reduce the likelihood of fracturing of the wire W. By employing such a configuration, the rebars R can be tied more tightly.

[0144] In one or more embodiments, the rebar tying tool **2** ties together rebars R using the wire W. The rebar tying tool **2** comprises: the reel **33** comprising the bobbin **160** and the wire W, which is wound on the bobbin **160**; the reel holder **10** (example of the reel-holding part) configured to hold the reel **33** in a rotatable manner; the feed part **38** configured to advance the wire W around the

rebars R from the reel 33; and the twisting part 46 configured to twist the wire W that is around the rebars R. The yield-point load per single wire W around the rebars R is 450 N or more.

[0145] If the wire W that ties together the rebars R deforms in an adverse manner, a gap or gaps will be formed between the rebars R and the wire W, and therefore the tying of the rebars R will loosen in an adverse manner. Consequently, to tie together the rebars R tightly, it is necessary to make it difficult for the wire W that ties together the rebars R to deform. Nevertheless, if the yield-point load per single wire W around the rebars R is small, if a strong force acts on the wire W that ties the rebars R, then there is a risk that the wire W will deform in an adverse manner. According to the above-mentioned configuration, because the yield-point load per single wire W around the rebars R is 450 N or more, even if a strong force acts on the wire W that ties together the rebars R, it is possible to reduce the likelihood of adverse deformation of the wire W. By employing such a configuration, the rebars R can be tied together more tightly.

[0146] Additional teachings, embodiments, structures, circuits, processes, methods, advantages, etc., which are combinable with the rebar tying tools described herein to develop additional embodiments of the present teachings, are disclosed in the Applicant's U.S. Pat. Nos. 11,529,669 and 11,554,409, the contents of which are fully incorporated herein by reference.

EXPLANATION OF THE REFERENCE NUMBERS

[0147] **2** Rebar tying tool [0148] **4** Main body [0149] **6** Grip [0150] **8** Battery-mount part [0151] **10** Reel holder [0152] **12** Trigger [0153] **14** Trigger switch [0154] **15** Support part [0155] **16** Housing [0156] **16a** Window [0157] **18** Right housing half [0158] **20** Left housing half [0159] **22** Motor cover [0160] **24** Operation-and-display part [0161] **24a** Main power-supply switch [0162] **24b** Main power-supply LED [0163] **26** Holder housing [0164] **26a** Pivot shaft [0165] **26b** Housing space [0166] **26c** Hole [0167] **28** Main cover [0168] **30** Auxiliary cover [0169] **30a** Auxiliary space [0170] **31** Torsion spring [0171] **32** Lock lever [0172] **33** Reel [0173] **36** Control circuit board [0174] **38** Feed part [0175] **40** Guide part [0176] **44** Cutting part [0177] **46** Twisting part [0178] **50** Feed motor [0179] **52** Speed-reducing part [0180] **54** Feed unit [0181] **56** Base member [0182] **58** Guide member [0183] **58a** Guide hole [0184] **60** Drive gear [0185] **62** First feed gear [0186] **62a** Groove [0187] **64** Second feed gear [0188] **64a** Groove [0189] **66** Release lever [0190] **66a** Pivot shaft [0191] **68** Compression spring [0192] **70** Upper-side curl guide [0193] **70a** Upper-side wire passageway [0194] **71** Lower-side curl guide [0195] **71a** Lower-side wire passageway [0196] **72** Fixed-cutter member [0197] **72a** Hole [0198] **74** Movable-cutter member [0199] **74a** Hole [0200] **76** First lever member [0201] **78** Second lever member [0202] **80** Link member [0203] **82** Torsion spring [0204] **86** Twisting motor [0205] **88** Speed-reducing part [0206] **90** Holding part [0207] **92** Rotation-blocking part [0208] **96** Bearing box [0209] **96a** Bearing [0210] **98** Carrier sleeve [0211] **98a** Clutch groove [0212] **98b** First wall part [0213] **98c** Second wall part [0214] **100** Clutch plate [0215] **100a** Clutch piece [0216] **102** Screw shaft [0217] **102a** Rear part [0218] **102b** Front part [0219] **102c** Flange [0220] **102d** Ball groove [0221] **102e** Engaging part [0222] **104** Inner sleeve [0223] **104a** Ball hole [0224] **104b** Flange [0225] **106** Outer sleeve [0226] **106a** Slit [0227] **108** Push plate [0228] **110** Clamp shaft [0229] **110a** Flat-plate part [0230] **110b** Opening [0231] **110c** Flange [0232] **110d** Hole [0233] **112** Right clamp [0234] **112a** Base part [0235] **112b** Downward protruding part [0236] **112c** Upward protruding part [0237] **112d** Contact part [0238] **112e** Upper-side guard part [0239] **112f** Front-side guard part [0240] **112g** Cam hole [0241] **112h** Cam hole [0242] **114** Left clamp [0243] **114a** Base part [0244] **114b** Pin-holding part [0245] **114c** Downward protruding part [0246] **114d** Contact part [0247] **114e** Rear-side guard part [0248] **114f** Front-side guard part [0249] **114g** Cam hole [0250] **114h** Cam hole [0251] **116** Compression spring [0252] **118** Compression spring [0253] **120** Ball [0254] **122** Pin [0255] **124** Cam sleeve [0256] **126** Cam sleeve [0257] **128** Support pin [0258] **130** Support pin [0259] **131** Cushion [0260] **132** Right-side wire passageway [0261] **134** Left-side wire passageway [0262] **138** Fin [0263] **138a** Short fin [0264] **138b** Long fin [0265] **140** Base member [0266] **140a** Pivot shaft [0267] **140b** Pivot shaft

Claims

1. A rebar tying tool configured to tie together rebars using a wire, comprising: a reel having the wire wound on a bobbin; a reel-holding part configured to hold the reel in a rotatable manner; a feed part configured to advance the wire from the reel by a number of turns of the wire around the rebars; and a twisting part configured to twist the wire after the number of turns of the wire has been wound around the rebars; wherein the rebar tying tool is configured such that the following two conditions are satisfied: the wire has a maximum tensile load of at least 1000 N; and the wire has a yield-point load of at least 550 N.
 2. The rebar tying tool according to claim 1, wherein the wire is composed of stainless steel.
 3. (canceled)
 4. The rebar tying tool according to claim 1, wherein: the twisting part comprises a tip-holding part configured to hold a tip of the wire that has been advanced around the rebars; and the feed part is configured to draw back the wire after the tip-holding part holds the tip of the wire and before the twisting part twists the wire.
 5. The rebar tying tool according to claim 1, wherein the wire has a diameter of at least 1.6 mm.
 6. The rebar tying tool according to claim 1, wherein a trunk part of the bobbin has a diameter of at least 50 mm.
 7. The rebar tying tool according to claim 1, wherein: the feed part comprises feed rollers configured to advance the wire by rotating; and the feed rollers have a hardness of at least 56 HRC.
 8. The rebar tying tool according to claim 7, further comprising: a cutting part comprising cutters configured to cut the wire; wherein the cutters have a hardness of at least 56 HRC.
 9. The rebar tying tool according to claim 1, further comprising: a grip configured to be grasped by a user during a rebar tying operation to perform the rebar tying operation in a hand-held state.
 10. The rebar tying tool according to claim 2, wherein: the twisting part comprises a tip-holding part configured to hold a tip of the wire that has been advanced around the rebars; and the feed part is configured to draw back the wire after the tip-holding part holds the tip of the wire and before the twisting part twists the wire.
 11. The rebar tying tool according to claim 10, wherein: the wire has a diameter of at least 1.6 mm; a trunk part of the bobbin has a diameter of at least 50 mm; the feed part comprises feed rollers configured to advance the wire by rotating, the feed rollers having a hardness of at least 56 HRC; and further comprising: a cutting part comprising cutters configured to cut the wire, the cutters having a hardness of at least 56 HRC; and a grip configured to be grasped by a user during a rebar tying operation to perform the rebar tying operation in a hand-held state.
 - 12.-19. (canceled)
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